

RESEARCH EXPOSÉ

Do Features Do the Work?

A Systematic Study of Feature Engineering and Model Architecture for Turbofan
Remaining Useful Life Prediction on the NASA C-MAPSS Dataset

Submitted by:	Matej Krsteski
Proposed supervisor:	Ass.-Prof. Patrick Link
Institution:	FH Technikum Wien
Date:	August 2026 (version 2)

Changes since version 1 (May 2026). Two references corrected: the comparative architecture study is by Goel et al. (2026), not Arkhipov, and the Alomari et al. article number is 13466. The feature counts in Experiment 1 are now measured rather than estimated, cycle transforms are separated into their own configuration, the PHM08 score is added to the metrics, every configuration is now repeated across three random seeds with differences tested by paired bootstrap, and the timeline is moved to reflect the actual start of experimental work.

1. Motivation

This research proposal grew directly out of a group project completed in the Data Science and Machine Learning course at FH Technikum Wien in the 2025/26 academic year. The project involved building an LSTM-based pipeline to predict the remaining useful life (RUL) of aircraft turbofan engines using the NASA C-MAPSS dataset. Our pipeline expanded 21 raw sensor channels into roughly 180 engineered temporal features, including rolling statistics, lag features, degradation trend slopes, and aggregated health metrics. The final model achieved RMSE values of 17.1, 23.7, 14.2, and 24.0 cycles across the four C-MAPSS subsets (FD001–FD004), within the range of results reported by baseline and mid-complexity approaches in the literature.

Those four figures come from a single unseeded run of a version of the pipeline that has since changed, and they are treated in this proposal as a historical reference point rather than as a baseline. Re-establishing a properly measured baseline is the first experiment described in Section 4, and the older configuration is retained as one of the anchor runs so that the difference between the two can be reported rather than assumed.

Prior to the project presentation, one question had already emerged as the central motivation behind this research proposal. Which components of our feature engineering pipeline actually contributed to predictive performance, and to what extent? Our final model relied on a large set of engineered features including rolling statistics, lag features, degradation trend slopes, and aggregated health metrics. However, we never evaluated the individual contribution of these feature groups. Instead, they were introduced cumulatively as a combined pipeline following the same general approach observed across much of the existing C-MAPSS literature. As far as I could find, no prior work has systematically isolated these feature engineering steps through an ablation study to determine whether certain transformations drive most of the accuracy gains while others are redundant or unnecessary.

During the project presentation, a second question emerged that further strengthened the relevance of this investigation. If the engineered features already have rich temporal data through rolling averages, lagged values, and degradation trends, does the LSTM's sequential processing still provide meaningful additional predictive power? Or could a simpler non-sequential model, such as a Random Forest operating only on the final engineered engine snapshot, achieve comparable performance?

Together, these questions form the foundation of this proposal.

2. Background and Problem Statement

Predictive maintenance and more specifically the task of predicting when a machine component will fail has become one of the most studied applied problems in machine learning. The NASA C-MAPSS dataset, introduced by Saxena et al. (2008), is the dominant benchmark for turbofan engine RUL prediction and has been used in hundreds of published studies over the past fifteen years.

Before going any further I must note that I haven't read every paper in this space since it's a large literature, but from the papers I have read I can highlight the following relevant papers.

A 2021 review by Vollert and Theissler surveyed the state of ML-based RUL prognosis on C-MAPSS specifically. Their key finding was that while the field has produced increasingly sophisticated model architectures, from basic LSTMs to hybrid CNN-LSTM-Attention networks, several fundamental challenges remain unaddressed. The review also noted that feature extraction methods are used inconsistently across studies, 43 of the 81 surveyed papers used no feature processing beyond normalisation, while the remainder employed varying approaches with no standard evaluation.

When you look at the C-MAPSS literature, almost every paper that achieves strong results uses some form of feature engineering, rolling windows, lag features, and similar transformations, but then proceeds to test only one feature configuration. The number of papers that go back and measure what would have happened without those features, or which specific engineering step contributed most to the improvement, are limited. In most papers, feature engineering is introduced as part of the preprocessing pipeline, but the individual contribution of those transformations is usually not evaluated separately. This is the gap this paper addresses.

A notable exception is the recent work by Alomari et al. (2023), who used aggregated feature importance methods to identify which features matter for RUL prediction on C-MAPSS. Their study pairs a TSFresh feature pool with four comparatively simple regressors, natural gradient boosting, extreme gradient boosting, Random Forest and a multi-layer perceptron, and reports RMSE of 11.8, 23, 14.6 and 22.3 across the four subsets, outperforming raw-sensor LSTM baselines on FD001, FD002 and FD004. This is a relevant finding, but their study focuses on feature selection after engineering, not on understanding which engineering steps are necessary in the first place.

There is also a parallel question about model architecture. The recent comparative study by Goel et al. (2026) tested Ridge Regression, Polynomial Ridge, XGBoost, a 1D CNN and an LSTM under an identical preprocessing pipeline on FD001 and FD003, and found that the LSTM achieved RMSE of 14.93 and 14.20 respectively. The LSTM was the strongest model on FD001, but on FD003 XGBoost reached 13.36 and beat it outright. This suggests that the gap between a well-engineered classical model and a temporal deep learning model may be smaller than typically assumed, and in at least one case runs the other way, thus raising the question of how much the LSTM's sequential processing contributes beyond what good feature engineering already accomplishes.

This motivates a clean, controlled experiment: hold the model fixed and vary the features, then hold the features fixed and vary the model. This paper proposes to do both.

3. Research Questions

This paper addresses two related questions:

RQ1 — Feature Engineering Ablation. Which feature engineering steps, that is rolling statistics, lag features, degradation trends, system state aggregates and cycle transforms, contribute meaningfully to LSTM-based RUL prediction accuracy on the NASA C-MAPSS dataset, and which are redundant?

RQ2 — Architecture versus Features. When the best-performing feature set from RQ1 is applied to non-sequential models operating on a single engineered snapshot, how does their accuracy compare to the LSTM operating on a 30-cycle sequence? Does temporal modelling add value beyond what explicit feature engineering already encodes?

These two questions are linked: RQ1 identifies the feature configuration carried forward, which RQ2 then uses as a controlled input. Together they answer the broader question of whether predictive power in this domain comes primarily from the features, the model architecture, or both.

4. Proposed Methodology

Dataset and evaluation

All experiments use the NASA C-MAPSS dataset across all four subsets (FD001–FD004). This allows conclusions to be tested against varying levels of complexity: FD001/FD003 have one operating

condition; FD002/FD004 have six. FD003/FD004 include two fault modes; FD001/FD002 include one. This variation lets us determine whether findings generalise across dataset complexity or are specific to simpler configurations.

The evaluation metrics are Root Mean Squared Error (RMSE) as the primary score, Mean Absolute Error (MAE), and the asymmetric PHM08 score introduced with the original data challenge, which penalises late predictions more heavily than early ones because that is the direction in which an error is dangerous. Training and prediction times are recorded for every run, since RQ2 is in part a question about whether additional model complexity is worth its cost. All metrics are reported per subset and averaged across all four. The piecewise-linear RUL cap (125 cycles for FD001/FD003, 150 for FD002/FD004) follows the approach validated during the course project and is consistent with recent literature.

Repetition and uncertainty. Every configuration is trained three times with different random seeds, and all reported figures carry the spread across seeds alongside the mean. Differences between configurations are tested with a paired bootstrap over test engines, drawing 10,000 resamples and treating a difference as real only when the central 95% interval excludes zero. Seed count rises to five if the observed spread exceeds 1.0 RMSE anywhere. This matters because the differences of interest in an ablation study are frequently of the same order as run-to-run variation, and a single unseeded run per cell cannot distinguish the two. The full experimental protocol is pre-registered in the project repository and committed before any results are collected.

Experiment 1 — Feature Engineering Ablation (RQ1)

The LSTM architecture is held fixed across all configurations (two-layer LSTM 128→64, Dropout 0.3, Dense 32, early stopping on a validation split held out by engine). Only the input features change. Six cumulative configurations are tested, each adding exactly one feature group to the one before it. The feature counts below are measured, not estimated, and differ across subsets because constant sensors are removed per subset.

Config	Feature set	FD001	FD002	FD003	FD004
C1_base	Settings and sensors only	17	19	18	19
C2_rolling	+ rolling mean and sd (windows 5, 10)	77	87	82	87
C3_lag	+ lag features (lags 1, 3)	107	121	114	121
C4_trend	+ degradation trend slopes (window 10)	122	138	130	138
C5_agg	+ system state aggregates	126	142	134	142
C6_full	+ cycle transforms	129	145	137	145

Cycle transforms are given their own configuration rather than being folded into a single "full pipeline" step. In the first version of this proposal they were bundled into Config 6, which would have made the final rung measure a different thing from what its label claimed.

Three further anchor runs are included so that the study can account for its own history. C6_full is the clean pipeline this paper reports. C6_legacy is the course-project code as it currently stands, which retains between 34 and 44 constant columns that carry no information. C6_expose is the version that produced the figures quoted in Section 1, and differs from C6_legacy by a single column: the k-means operating-condition index supplied to the network as an ordinary numeric feature. On FD001 and

FD003, which have one real operating condition and settings that vary by roughly 0.002, that index is grouping measurement noise. Running all three costs little and allows the paper to state, with evidence, what cleaning the feature set actually changed.

Each configuration is run on all four subsets across three seeds, giving 72 runs for the ladder and a further 24 for the anchors. A secondary leave-one-out experiment on FD001 and FD004 removes each feature group from the full set in turn, adding 15 runs. The ladder measures what a group adds given the groups before it, while leave-one-out measures what it adds given everything else; the two can disagree, and where they do, the disagreement is reported rather than resolved in favour of the more convenient reading.

Experiment 2 — Architecture Comparison (RQ2)

Using the feature configuration carried forward from Experiment 1, chosen by a selection rule fixed in advance, four models are compared:

Model	Input	Library	Key question
A — LSTM	30-cycle sequences $\times$ N features	TensorFlow / Keras	Existing baseline
B — RF	Single row (last cycle) $\times$ N features	scikit-learn, 500 trees	Is temporal processing redundant?
C — DT	Single row (last cycle) $\times$ N features	scikit-learn, CV depth	Interpretable lower bound
D — RF flat	All 30 cycles flattened $\times$ N features	scikit-learn, 500 trees	Order, or simply more numbers?

The critical design constraint is that Models B, C and D receive the same engineered features as Model A, including rolling statistics and trend slopes that already encode recent history mathematically. Model D is what separates the two competing explanations for any LSTM advantage. If the LSTM outperforms Model B, that could mean sequence order carries information, or merely that the LSTM was shown thirty times as many numbers. Model D sees exactly the same numbers as the LSTM with no notion of their order, so comparing the three settles the question: if D matches A, the architecture adds nothing the features had not already supplied; if D falls short of A, reading in order genuinely helps; and if B matches D, the final cycle already contained everything.

Experiment 2 requires no additional deep-learning runs, since its LSTM arm is a configuration already trained in Experiment 1. The tree-based models train in minutes.

5. Expected Contributions and Significance

This study would make three concrete contributions:

1. **A systematic feature engineering ablation study on the full C-MAPSS benchmark (all four subsets).** To the best of the author's knowledge, no published work has tested incrementally constructed feature sets under a controlled model, with repeated seeds and tested differences, to isolate the contribution of each engineering step. The closest related work (Alomari et al., 2023) uses post-hoc feature importance rather than pre-hoc ablation, and does not separate the contribution of individual engineering categories.

2. **A direct empirical answer to whether the LSTM's temporal processing provides accuracy gains beyond explicit feature engineering.** This answer would help practitioners decide whether complex sequential models are warranted or whether simpler models with richer features suffice, and the flattened-window control makes the answer a three-way one rather than a binary.
3. **A reproducible, publicly available pipeline.** All code, the pre-registered protocol, and one logged line per training run are public from the start of the work rather than at the end of it. Given that C-MAPSS is the dominant benchmark in the field, a clean and well-documented ablation pipeline would serve as a useful baseline resource for future work.

None of these contributions are technically novel, rather their value is in doing a clean comparison and documenting it, which is currently missing in the field.

6. Possible Venues

Primary targets, domain-focused journals:

- Reliability Engineering & System Safety (Elsevier). Publishes applied predictive maintenance and prognostics research and a lot of C-MAPSS work, and would be a natural fit for an ablation study.
- Mechanical Systems and Signal Processing (Elsevier). Regularly publishes C-MAPSS-based RUL papers, and the ablation framing would be novel in this venue.

Secondary targets, conferences and workshops:

- PHM Society Annual Conference. The primary community working with C-MAPSS, and the venue where the dataset itself was introduced.
- IEEE ETFA (Emerging Technologies and Factory Automation). The Vollert and Theissler (2021) review was published here, making it a directly relevant venue for a follow-up study.
- Applied Machine Learning workshops at major venues (ICML, ECML-PKDD), if framed as a contribution to empirical ML methodology rather than the domain application.

The final venue selection should be made after the supervisor completes their review of the landscape.

7. Proposed Timeline

Period	Activity
August 2026	Environment restored and baseline re-measured. Experimental protocol pre-registered and committed. Config-driven feature builder implemented and feature counts verified. Pilot run to measure seed-to-seed variation and run time.
August – September 2026	Run the full grid: 96 training runs for Experiment 1 and its anchors, plus the leave-one-out check and the tree-based arms of Experiment 2. Generate tables, figures and significance tests.
September 2026	Draft the setup and results sections and share with supervisor for feedback.
September – October 2026	Write introduction, related work and discussion. Revise per supervisor feedback.

Period	Activity
October – November 2026	Finalise and submit to target venue.

Most of the code already exists. All experiments run locally on consumer hardware in an estimated eight to twelve hours of compute, so no external compute resources are required.

8. References

- Alomari, Y., Andó, M., & Baptista, M. L. (2023). Advancing aircraft engine RUL predictions: an interpretable integrated approach of feature engineering and aggregated feature importance. *Scientific Reports*, 13, 13466. <https://doi.org/10.1038/s41598-023-40315-1>
- Alomari, Y., Andó, M., & Baptista, M. L. (2024). Author Correction: Advancing aircraft engine RUL predictions: an interpretable integrated approach of feature engineering and aggregated feature importance. *Scientific Reports*, 14, 15726. <https://doi.org/10.1038/s41598-024-66604-x>
- Goel, A., Galchar, S., & Kanu, S. (2026). Remaining Useful Life Estimation for Turbofan Engines: A Comparative Study of Classical, CNN, and LSTM Approaches. *arXiv:2604.27234* [preprint, not peer-reviewed].
- Saxena, A., Goebel, K., Simon, D., & Eklund, N. (2008). Damage propagation modeling for aircraft engine run-to-failure simulation. In *2008 International Conference on Prognostics and Health Management*. IEEE.
- Vollert, S., & Theissler, A. (2021). Challenges of machine learning-based RUL prognosis: A review on NASA's C-MAPSS data set. In *26th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, Västerås, Sweden. IEEE.